# BEACH SIMULATOR

# Game design insertion map

Where each mechanic can be introduced, how modules can communicate, and which boundaries keep the code simple enough for people to work with.

| | |
|---|---|
| CODE BASE | Local architecture validated on July 21, 2026 |
| DESIGN BASE | Jesse Schell - The Art of Game Design: A Book of Lenses |
| CORE RULE | Add rules where they belong. Create a new script only when a new and lasting responsibility actually exists. |

# 1. How to use this map

This document is not permission to build any feature anywhere. It is a decision map: first identify the nature of the mechanic, then change the module that already owns that state or rule.

## Rule of thumb

• If a feature changes an existing rule, modify the model that already controls that rule.
• If it changes appearance only, modify the corresponding view or visual module.
• If it schedules something for later, use the spawn timeline or the day lifecycle.
• If it crosses two domains, return a small result and route it through the composition point.
• Create another file only when the concept has its own name, state, and a realistic expectation of growth.

## Signs of overengineering

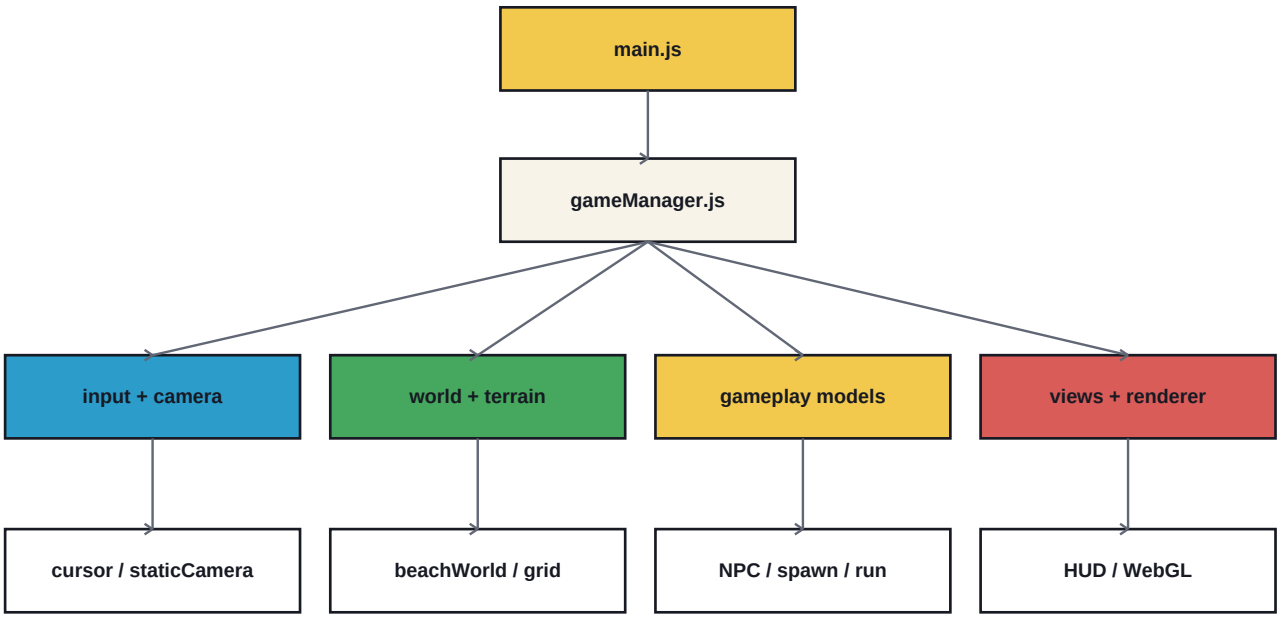
• A new file contains only one call to another file.
• Four scripts must be opened to understand one click.
• A generic Manager knows about camera, UI, NPCs, and money at the same time.
• Events are created for calls that could remain direct and local.
• DTOs, registries, and factories exist without more than one real implementation.

## Reading the lenses

The lenses below are used as design questions, not as software patterns. A lens explains why a mechanic may improve the experience; the listed module explains where to implement it without contaminating the architecture.

# 2. Resulting architecture

The dependency direction remains top-down. **gameManager.js** composes systems and owns the loop. Lower modules never import it. Concrete beach assembly now lives in **world/beachWorld.js**, reducing GameManager without introducing an orchestration framework.




## Interpretation

• main.js only starts the game.
• gameManager.js connects input, loop, models, and presentation.
• beachWorld.js assembles terrain, scenery, grid, initial population, and assets.
• Gameplay models do not import the DOM, camera, or renderer.
• Views receive snapshots or presentation commands; they do not decide economy or behavior.

# 3. Quick map

Use this table to quickly locate the most likely insertion point for a change.

| SCRIPT | DESIGN DECISION | EXAMPLE |
|---|---|---|
| world/beachWorld.js | Initial map composition | Add a static building, initial position, or reserved area |
| terrain/beachGrid.js | Playable space and occupancy | Collision, store footprint, blocked cells |
| terrain/terrainWorld.js | Terrain shape and visuals | Wet sand, vegetation strip, palm density |
| spawn/spawnManager.js | When something happens | Bather arrival, shark, litter, or treasure |
| spawn/spawnableObjectDto.js | What may exist | New litter, animal, valuable, or obstacle |
| npcs/npcSystem.js | What an NPC decides | Needs, conversation, swimming, fleeing, and leaving |
| buildings/buildingServicesModel.js | Service effect and cost | Items, prices, wages, and maintenance |
| interaction/cleanBeachController.js | Collection consequence | Money, progress, tools, and combos |
| onboarding/cleanBeachGuide.js | Contextual first-task help | Delay, target choice, and arrow priority |
| run/dayLifecycleController.js | Day rhythm | Climax, closing, charges, and spawn opening |
| camera/staticCamera.js | Camera state and movement | Focus, limits, and minimap data |
| input/cursor.js | Raw gestures | Touch, drag, zoom, and rotation |
| ui/*.js | How state is shown | HUD, report, tooltip, notice, and animation |

# 4. World, terrain, and space

| world/beachWorld.js | |
|---|---|
| CURRENT RESPONSIBILITY | Loads assets and assembles the initial beach: terrain, kiosk, beach house, grid, an empty NPC set, and five collection objects. |
| GAME DESIGN INSERTIONS | • Balance the five initial objects and guaranteed valuable in planInitialPopulation.<br>• Position new buildings that already exist at the start of a run.<br>• Reserve footprints so objects and NPCs cannot appear inside buildings.<br>• Define the initial visual composition that creates the run's first impression. |
| COMMUNICATION NODE | planInitialPopulation returns DTO requests; loadBeachWorld returns concrete references and resolveWorldObjectPosition to GameManager. |
| DO NOT PUT HERE | Prices, maintenance, NPC behavior, timers, or reward rules. |
| BOOK FOUNDATION | **Lens #21 Functional Space (p. 135); #62 Inherent Interest (p. 254); #72 Indirect Control (p. 293).** |

| terrain/beachGrid.js | |
|---|---|
| CURRENT RESPONSIBILITY | Divides the map into sea, sand, and vegetation zones; converts positions into cells and tracks occupancy. |
| GAME DESIGN INSERTIONS | • Building footprints with different dimensions.<br>• An isWalkable query for simple bather collision.<br>• Navigation costs for dry sand, wet sand, and shallow water. |
| COMMUNICATION NODE | NPC placement and object placement can query the grid through an injected function. The grid must not know specific NPCs or buildings. |
| DO NOT PUT HERE | Animation, money, timed spawns, or renderer access. |
| BOOK FOUNDATION | **Lens #21 Functional Space (p. 135); #22 Dynamic State (p. 140); #42 Simplicity/Complexity (p. 196).** |

| terrain/terrainWorld.js | |
|---|---|
| CURRENT RESPONSIBILITY | Builds and updates the visual sand, vegetation, sea, and palm cells around the camera. |
| GAME DESIGN INSERTIONS | • A wet-sand strip or visual variation by zone.<br>• Palm distribution, scale, and spacing.<br>• Purely visual decoration and the terrain's infinite window. |
| COMMUNICATION NODE | May receive simple visual state, such as tide intensity, without importing TimeManager. |
| DO NOT PUT HERE | Collision, land price, needs, or spawn rules. |
| BOOK FOUNDATION | **Lens #7 Elemental Tetrad (p. 43); #63 Beauty (p. 255); #72 Indirect Control (p. 293).** |

# 5. Spawning and objects

| spawn/spawnManager.js | |
|---|---|
| CURRENT RESPONSIBILITY | Schedules events, controls arrival intervals, and turns bathers into future litter sources. |
| GAME DESIGN INSERTIONS | • New channels: shark, crab, jellyfish, customer rush, or treasure.<br>• Probability, cooldown, priority, and attention cost for each event.<br>• Adaptive pacing using rating and, later, a summary of the player's behavioral DNA. |
| COMMUNICATION NODE | Returns spawn requests. It does not instantiate 3D models or directly change the HUD or economy. |
| DO NOT PUT HERE | WebGL position, animation, charges, or NPC dialogue. |
| BOOK FOUNDATION | **Lens #2 Surprise (p. 26); #29 Chance (p. 169); #39 Time (p. 189); #61 Interest Curve (p. 252).** |

| spawn/scheduledSpawnQueue.js | |
|---|---|
| CURRENT RESPONSIBILITY | Maintains the time-ordered queue of future events. |
| GAME DESIGN INSERTIONS | • Only technical tie-breaking criteria required by scheduling.<br>• Cancellation support when an event's gameplay condition disappears. |
| COMMUNICATION NODE | Receives event DTOs from SpawnManager and returns the next event. It must remain unaware of beaches, NPCs, or sharks. |
| DO NOT PUT HERE | Probabilities, narrative, rewards, or event-type selection. |
| BOOK FOUNDATION | **Lens #39 Time (p. 189); #43 Elegance (p. 198).** |

| spawn/spawnableObjectDto.js | |
|---|---|
| CURRENT RESPONSIBILITY | Universal catalog of object types, categories, zones, sources, and traits that may exist in the world. |
| GAME DESIGN INSERTIONS | • Register a coconut, jewel, rock, log, animal, or new type of litter.<br>• Declare reusable traits such as dirty, pickup, obstacle, or predator.<br>• Define the zones and sources where an object is allowed. |
| COMMUNICATION NODE | Other modules query the DTO by type. The DTO describes; it does not execute behavior. |
| DO NOT PUT HERE | Spawn timing, prices, animation, or concrete effects on NPCs. |
| BOOK FOUNDATION | **Lens #22 Dynamic State (p. 140); #23 Emergence (p. 143); #26 Rules (p. 150).** |

# 6. NPCs and buildings

| npcs/npcSystem.js | |
|---|---|
| CURRENT RESPONSIBILITY | Maintains entities, movement, acceleration, activities, needs, complaints, and departures from the beach. |
| GAME DESIGN INSERTIONS | • Interaction between two bathers: conversation, friendship, argument, or family group.<br>• Name, age, hygiene profile, heat tolerance, and final review.<br>• States for swimming, fleeing an animal, buying a drink, and seeking a toilet. |
| COMMUNICATION NODE | update may return small events such as departures and interactions. GameManager routes those results to rating, economy, or UI. |
| DO NOT PUT HERE | DOM, visual speech bubbles, GLTF loading, money, or global scheduling. |
| BOOK FOUNDATION | **Lens #16 Player (p. 106); #19 Needs (p. 127); #23 Emergence (p. 143); #44 Character (p. 199).** |

| npcs/npcWorld.js | |
|---|---|
| CURRENT RESPONSIBILITY | Loads the PicoCAD/GLTF model and converts NPC snapshots into renderable bodies and legs. |
| GAME DESIGN INSERTIONS | • New animations for swimming, complaining, buying, and talking.<br>• Visual bather variations that preserve the same behavior.<br>• Explicit orientation offsets for each new model. |
| COMMUNICATION NODE | Receives ready NPC snapshots from NpcSystem. It does not decide where or why an NPC moves. |
| DO NOT PUT HERE | Needs, pathfinding, ratings, prices, or departure rules. |
| BOOK FOUNDATION | **Lens #7 Elemental Tetrad (p. 43); #44 Character (p. 199); #57 Feedback (p. 230).** |

# 6. NPCs and buildings - services

| buildings/buildingServicesModel.js | |
|---|---|
| CURRENT RESPONSIBILITY | Tracks owned buildings and calculates their effects, maintenance, and end-of-day costs. |
| GAME DESIGN INSERTIONS | • Beverage Store items and prices while the catalog remains small.<br>• Stock, margin, lifeguard wages, and toilet maintenance cost.<br>• Clear trade-offs: income versus litter, safety versus wages, hygiene versus maintenance. |
| COMMUNICATION NODE | Receives commands such as addBuilding or purchase. Returns a financial result; the economy records the transaction. |
| DO NOT PUT HERE | 3D models, choice modal, NPC movement, or temporal event ordering. |
| BOOK FOUNDATION | **Lens #32 Meaningful Choices (p. 181); #33 Triangularity (p. 182); #46 Economy (p. 204); #47 Balance (p. 205).** |

| scenery/sceneryWorld.js | |
|---|---|
| CURRENT RESPONSIBILITY | Defines assets, scale, orientation, and visual instances for the kiosk, beach house, and placeholders. |
| GAME DESIGN INSERTIONS | • Register the visual asset for toilets, trash cans, Wi-Fi spot, and beverage store.<br>• Correct model-specific yaw, size, and brightness. |
| COMMUNICATION NODE | Receives type and position from world composition or the building choice. |
| DO NOT PUT HERE | Prices, service effects, maintenance, or random run selection. |
| BOOK FOUNDATION | **Lens #7 Elemental Tetrad (p. 43); #63 Beauty (p. 255).** |

# 7. Interaction, input, and camera

| interaction/cleanBeachController.js | |
|---|---|
| CURRENT RESPONSIBILITY | Applies collection consequences: money, experience, and task progress. |
| GAME DESIGN INSERTIONS | • Different values for coins, rings, and jewels.<br>• A tool that removes a specific hazard.<br>• A combo or bonus for rapid cleanup streaks, if playtesting supports it. |
| COMMUNICATION NODE | Receives the collected object and calls only the involved models. Returns a small result for visual feedback. |
| DO NOT PUT HERE | Hit testing, DOM, animation, camera, or object creation. |
| BOOK FOUNDATION | **Lens #40 Reward (p. 191); #49 Visible Progress (p. 214); #57 Feedback (p. 230); #58 Juiciness (p. 233).** |

| interaction/worldObjectInteraction.js | |
|---|---|
| CURRENT RESPONSIBILITY | Projects 3D objects onto the canvas or HUD and resolves hover or click through pixel-based trigger areas. |
| GAME DESIGN INSERTIONS | • Box selection, building drag, or priority between overlapping objects.<br>• Category-specific hit-area tuning for accessibility.<br>• Positioning new speech bubbles or indicators attached to world entities. |
| COMMUNICATION NODE | Returns a selection or projected coordinates. A gameplay controller decides the effect of the click. |
| DO NOT PUT HERE | Money, tasks, inventory removal, or NPC rules. |
| BOOK FOUNDATION | **Lens #48 Accessibility (p. 213); #53 Control (p. 222); #55 Virtual Interface (p. 226).** |

| input/cursor.js | |
|---|---|
| CURRENT RESPONSIBILITY | Normalizes mouse and keyboard into pan, rotation, zoom, selection, and edge-scrolling intents. |
| GAME DESIGN INSERTIONS | • Mobile mapping for pinch, tap, one-finger drag, and two-finger rotation.<br>• Sensitivity and direction-inversion settings. |
| COMMUNICATION NODE | Emits input intents. It never calls NPC, terrain, or GameManager directly. |
| DO NOT PUT HERE | Rewards, collision, spawning, concrete camera state, or world mutation. |
| BOOK FOUNDATION | **Lens #35 Head and Hands (p. 185); #53 Control (p. 222); #54 Physical Interface (p. 226).** |

# 7. Interaction, input, and camera - camera

| camera/staticCamera.js | |
|---|---|
| CURRENT RESPONSIBILITY | Maintains the RTS camera target, distance, yaw, pan, rotation, and zoom. |
| GAME DESIGN INSERTIONS | • Focus on an NPC, building, or urgent event.<br>• Navigation limits and a smooth return to gameplay.<br>• Provide target and bounds for a future minimap. |
| COMMUNICATION NODE | For a minimap, the composition point reads camera.getTarget and sends a snapshot to MinimapView. The camera does not import the view. |
| DO NOT PUT HERE | DOM, minimap icons, day rules, spawning, or money. |
| BOOK FOUNDATION | **Lens #53 Control (p. 222); #56 Transparency (p. 227); #64 Projection (p. 257).** |

# 8. Run, time, and metrics

| run/dayLifecycleController.js | |
|---|---|
| CURRENT RESPONSIBILITY | Starts the day, enables spawning, ends the day, stops the timeline, and records service costs. |
| GAME DESIGN INSERTIONS | • Trigger a climax during the last forty seconds.<br>• Prepare the final summary and transition to the next day.<br>• Apply run modifiers before spawning is enabled. |
| COMMUNICATION NODE | Coordinates day-cycle models, but does not assemble UI or decide individual NPC behavior. |
| DO NOT PUT HERE | Rendering, camera, assets, dialogue, or a building's internal rules. |
| BOOK FOUNDATION | **Lens #39 Time (p. 189); #60 Modes (p. 240); #61 Interest Curve (p. 252).** |

| run/runSessionModel.js | |
|---|---|
| CURRENT RESPONSIBILITY | Stores the current day, total days, and current run phase. |
| GAME DESIGN INSERTIONS | • Progression from Day 1 through Day 5.<br>• States for day-intro, active, day-complete, and run-complete.<br>• Run victory or defeat conditions after they are approved. |
| COMMUNICATION NODE | The lifecycle requests transitions; views observe snapshots routed by the composition point. |
| DO NOT PUT HERE | Per-frame countdown, UI, charges, or spawning. |
| BOOK FOUNDATION | **Lens #25 Goals (p. 149); #39 Time (p. 189); #60 Modes (p. 240).** |

| time/timeManager.js | |
|---|---|
| CURRENT RESPONSIBILITY | Controls day duration, remaining time, start, and clock updates. |
| GAME DESIGN INSERTIONS | • Different duration by difficulty or run modifier.<br>• Explicit pause during modals, if the design adopts that rule.<br>• A final-window signal for the day's climax. |
| COMMUNICATION NODE | Returns a time snapshot. Other modules choose how to react to time. |
| DO NOT PUT HERE | Concrete spawns, wages, tasks, animation, or clock text. |
| BOOK FOUNDATION | **Lens #18 Flow (p. 122); #39 Time (p. 189); #61 Interest Curve (p. 252).** |

| economy/beachEconomyModel.js | |
|---|---|
| CURRENT RESPONSIBILITY | Maintains the balance and a minimal history of income and expenses in cents. |
| GAME DESIGN INSERTIONS | • Transaction categories for the end-of-day report.<br>• Cash or debt limits only if they become an approved rule. |
| COMMUNICATION NODE | Services and controllers send transactions with sourceId. Economy does not ask why the purchase happened. |
| DO NOT PUT HERE | Item prices, sale probability, UI, or customer behavior. |
| BOOK FOUNDATION | **Lens #5 Endogenous Value (p. 32); #40 Reward (p. 191); #46 Economy (p. 204).** |

# 8. Run, time, and metrics - reviews

| ratings/beachRatingModel.js | |
|---|---|
| CURRENT RESPONSIBILITY | Stores reviews and calculates the average used by the HUD and spawn frequency. |
| GAME DESIGN INSERTIONS | • Bather departure review based on cleanliness, fulfilled needs, and safety.<br>• Review weighting or recency after playtesting.<br>• A summary of positive and negative reasons at day close. |
| COMMUNICATION NODE | NpcSystem returns departures; a controller converts the final experience into a rating and calls submitRating. |
| DO NOT PUT HERE | Move NPCs, draw stars, directly alter spawning, or decide animation. |
| BOOK FOUNDATION | **Lens #20 Judgment (p. 128); #40 Reward (p. 191); #57 Feedback (p. 230).** |

| experience/playerExperienceModel.js | |
|---|---|
| CURRENT RESPONSIBILITY | Accumulates signals of mastery, pressure, exploration, fatigue, and need for guidance. |
| GAME DESIGN INSERTIONS | • Adjust only pacing, hints, or future event selection.<br>• Detect a lost player without punishment or hidden reward reduction. |
| COMMUNICATION NODE | Exposes a compact snapshot to SpawnManager or onboarding. It does not receive DOM or directly control the timeline. |
| DO NOT PUT HERE | Decide financial results, manipulate ratings, or replace clear rules with invisible difficulty. |
| BOOK FOUNDATION | **Lens #16 Player (p. 106); #18 Flow (p. 122); #31 Challenge (p. 179).** |

# 9. UI, feedback, and rendering

| ui/buildingChoice.js | |
|---|---|
| CURRENT RESPONSIBILITY | Maintains the current roguelike choice and presents three options to the player. |
| GAME DESIGN INSERTIONS | • Show the benefit, daily cost, and risk of each building.<br>• Ensure the selected consequence appears within the same minute.<br>• Vary options by run without creating one dominant strategy. |
| COMMUNICATION NODE | The view returns only the selected type. BuildingServices applies the rule; scenery creates the visual. |
| DO NOT PUT HERE | Charge money, spawn NPCs, or directly alter terrain. |
| BOOK FOUNDATION | **Lens #32 Meaningful Choices (p. 181); #33 Triangularity (p. 182); #40 Reward (p. 191).** |

| ui/taskList.js | |
|---|---|
| CURRENT RESPONSIBILITY | Maintains and presents tasks, priority, dependencies, and progress. |
| GAME DESIGN INSERTIONS | • Concrete short-term goals with a counter and bar.<br>• Conditional tasks for cleanliness, satisfaction, or safety.<br>• Reveal the next task only when the player has enough context. |
| COMMUNICATION NODE | Controllers advance tasks by id. The view reacts to snapshots and does not query the world. |
| DO NOT PUT HERE | Remove objects, pay rewards, or decide spawning. |
| BOOK FOUNDATION | **Lens #25 Goals (p. 149); #49 Visible Progress (p. 214); #57 Feedback (p. 230).** |

| onboarding/cleanBeachGuide.js + feedback views | |
|---|---|
| CURRENT RESPONSIBILITY | The guide selects a pending item after inactivity; OnboardingView and PickupFeedbackView present the hand, arrow, notices, +1, and $5. |
| GAME DESIGN INSERTIONS | • Tune the 3-to-5-second window using playtest results.<br>• Prioritize visible, valuable, or distant objects according to tutorial intent.<br>• Add a second contextual hint without placing the rule inside the view. |
| COMMUNICATION NODE | CleanBeachGuide queries task and objects, chooses the target, and sends projected coordinates. Views only draw the feedback. |
| DO NOT PUT HERE | Change money, remove objects, complete tasks, or control the camera. |
| BOOK FOUNDATION | **Lens #48 Accessibility (p. 213); #57 Feedback (p. 230); #58 Juiciness (p. 233).** |

| rendering/worldRenderer.js | |
|---|---|
| CURRENT RESPONSIBILITY | Executes WebGL, shaders, sprites, lighting, and the final frame at a fixed internal resolution. |
| GAME DESIGN INSERTIONS | • PSX shader, jittering, water, transparency, fog, and graphics optimization.<br>• New visual passes only when an effect cannot fit the current material. |
| COMMUNICATION NODE | Receives ready sceneObjects. It does not query GameManager or domain models. |
| DO NOT PUT HERE | Ratings, spawning, selection, prices, collision, or NPC behavior. |
| BOOK FOUNDATION | **Lens #7 Elemental Tetrad (p. 43); #63 Beauty (p. 255); #64 Projection (p. 257).** |

# 10. Three insertion examples

## A. Store items and prices

• buildings/buildingServicesModel.js: a small list of beverages, prices, margins, and stock.
• npcs/npcSystem.js: BUYING state and the decision to seek the store.
• economy/beachEconomyModel.js: records only the already-calculated sale.
• scenery/sceneryWorld.js: store asset and visual orientation.
• ui/buildingChoice.js: communicates the trade-off before selection.

## B. Two bathers interact

• npcs/npcSystem.js detects proximity and creates an interaction result.
• The same system changes both NPC states to TALKING or ARGUING.
• npcs/npcWorld.js animates both bodies.
• A view receives dialogue only when text actually exists.
• SpawnManager does not participate: the interaction comes from current state, not a global clock.

## C. Opening a minimap communication node

• camera/staticCamera.js already provides target and orientation.
• terrain/beachGrid.js provides zones and logical boundaries.
• NpcSystem and world objects provide minimal position snapshots.
• A future MinimapView receives one simple object containing those values.
• GameManager only assembles the snapshot during render; camera, NPC, and grid do not import the view.

## Suggested contract, only after the minimap is approved

```
minimapView.render({ cameraTarget, cameraYaw, zones, npcs, objects })
```

# 11. Recommended backlog

Suggested order for improving the experience without concentrating rules in GameManager again.

| SCRIPT | DESIGN DECISION | EXAMPLE |
|---|---|---|
| 1. Five days | Close the core loop | advanceDay, clock reset, and run-complete |
| 2. NPC collision | Preserve spatial readability | Inject beachGrid.isWalkable into NpcSystem |
| 3. Meaningful buildings | Make choices matter | Visible trade-offs and consequences within the same minute |
| 4. First customer | Create attachment and curiosity | Name, need, and first review |
| 5. Complete rating loop | Close systemic feedback | Departure creates a review and changes future frequency |
| 6. Climax and report | Shape the five minutes | Final event, balance, rating, cleanup, and teaser |
| 7. Emergent events | Increase variety | NPC interactions, animals, litter, and treasure |

## Criterion for creating a new script

Create a file only when at least two conditions are true: it has its own state; it has its own rule; it has more than one consumer; the concept deserves its own tests; the current file would become difficult to understand with the feature. Otherwise, modify the existing method or module.

## Priority lenses for the next five minutes

- #25 Goals: the player always knows what to do now.
- #32 Meaningful Choices: every building changes the run.
- #39 Time: events fit the five-minute rhythm.
- #49 Visible Progress: cleanliness and satisfaction have concrete evidence.
- #57 Feedback and #58 Juiciness: every action responds immediately.
- #61 Interest Curve: hook, rest, escalation, climax, and resolution.